DELA AKPALU

Quantitative Researcher | Statistician | Data Scientist

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Profile

Quantitative researcher with graduate training in statistical science and data science, combining statistical inference, time-series modeling, machine learning, and reproducible software workflows. Experienced in Python, R, SQL, neural networks, VAR models, anomaly detection, causal inference, and research data systems. Builds transparent evaluation pipelines, tests model robustness, and communicates technical results to research, operational, and instructional audiences.

Technical Skills

Programming: Python, R, SQL, C++, Julia, Git, LaTeX
Statistical Methods: Probability, statistical inference, regression, Bayesian statistics, causal inference, survival analysis, experimental design, stochastic processes
Machine Learning: scikit-learn, PyTorch, classification, anomaly detection, feature engineering, neural networks, model selection, precision-recall analysis
Time Series and Data Systems: VAR, recurrent neural networks, nowcasting, temporal validation, Spark, MySQL, REDCap, reproducible data pipelines
Visualization and Delivery: Matplotlib, Tableau, Power BI, technical reporting, dashboard design, Jira

Selected Projects

Stock-Market Nowcasting with Recurrent Models *Python, PyTorch, GRU, time series*
Project repository
- Built an end-to-end pipeline that generates market series, constructs a 13-variable rolling information set, trains a GRU model, and saves models, predictions, and structured metrics.
- Achieved validation MAE of 0.00962 across 313 observations and evaluation MAE of 0.01762 across 511 observations on normalized, generated market data; documented the deterioration outside the validation window.
- Separated data generation, feature construction, training, and evaluation to support repeatable experiments and walk-forward extensions.

Credit-Card Fraud Detection Benchmark *Python, scikit-learn, imbalanced learning*
Project repository
- Corrected and benchmarked logistic regression, random forest, and Isolation Forest pipelines on 221,409 training and 56,277 development transactions after duplicate removal.
- Selected random forest by average precision (0.867); obtained ROC AUC 0.961, precision 0.937, recall 0.763, and F1 0.841 on the fixed development split.
- Added class weighting, pipeline-contained scaling, deterministic seeds, model persistence, and machine-readable evaluation outputs.

Human Activity Classification *Python, scikit-learn, synthetic sensors*
Project repository
- Created a reproducible five-class wearable-sensor simulation and a stratified random-forest benchmark with saved data, confusion matrix, classification report, and metrics.
- Reached 84.4% held-out accuracy and 0.844 macro-F1 across 1,500 test windows versus a 20.0% most-frequent baseline; documented limits of simulator-based validation.

Logistics Recommender Evaluation *Python, ranking metrics, synthetic interactions*
Project repository
- Built a seeded interaction generator and leave-last-interaction-out evaluation for popularity and item co-occurrence recommenders across 320 users and 80 items.
- Improved Hit Rate@5 from 5.00% to 9.38% and MRR@5 from 0.0316 to 0.0580 in the educational simulation.

Research and Professional Experience

Analytics and Impact Lead **2026–Present**
Greater Faith Deliverance Ministries / Planning Committee / YAYA Tutoring *Project-Based / Volunteer*

- Develop a KPI and analytics framework for program outcomes, volunteer capacity, and operational progress across 21 auxiliaries.
- Support data collection, dashboarding, STEM pilot evaluation, survey design, engagement measurement, and impact reporting.
- Configure Jira workflows to track initiatives, owners, deadlines, blockers, and quarterly progress.

Research Assistant **Jun 2024–May 2025**
Wake Forest University *Winston-Salem, NC*

- Improved REDCap data quality and validation to support reliable storage, retrieval, and analysis of clinical research datasets.
- Automated data-cleaning and validation workflows in R, reducing manual processing and strengthening reproducibility.
- Collaborated with research teams to structure, maintain, and prepare large datasets for statistical analysis and reporting.

Graduate Teaching Assistant **Aug 2023–May 2025**
Wake Forest University *Winston-Salem, NC*

- Taught and supported statistics, inference, exploratory analysis, visualization, and reproducible R/Python workflows.
- Received the 2025 Department of Statistical Sciences Distinguished Teaching Assistant Award.

Mathematics Teacher **Nov 2025–Present**
St. Vincent Academy *Newark, NJ*

- Teach Algebra II, Geometry, and Calculus, emphasizing mathematical reasoning, quantitative fluency, and clear problem-solving.
- Use assessment data to identify learning gaps and develop targeted LaTeX-based notes, assessments, and worked examples.

Teaching Assistant **Sep 2020–Aug 2022**
Kwame Nkrumah University of Science and Technology *Kumasi, Ghana*

- Led instruction in experimental design, algebra, calculus, and statistical methods; mentored students on research and data projects.

Education

M.S., Statistical Science **Aug 2023–May 2025**
Wake Forest University *Winston-Salem, NC*
Causal Inference, Data Mining, Bayesian Statistics, Machine Learning, Survival Analysis

Structured M.Sc., Data Science **Aug 2022–Jun 2023**
African Institute for Mathematical Sciences *Accra, Ghana*
Big Data Analytics, Network Science, High-Dimensional Data Analysis

B.S., Statistics **Sep 2016–Aug 2020**
Kwame Nkrumah University of Science and Technology *Kumasi, Ghana*
Regression Analysis, Demographic Statistics, Stochastic Processes

Publication, Award, and Certification

- **Coauthor, Frontiers in Aging (2025).** “Bone mineral density and turnover response to GLP-1 receptor agonists in older adults with overweight/obesity and prediabetes/type 2 diabetes: a 20-week pilot trial post hoc analysis.”
- **Distinguished Teaching Assistant Award,** Wake Forest University Department of Statistical Sciences, 2025.
- **Data Science Certificate,** Wake Forest University, 2025.